Candidate ID No:……………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. Examination (CAD/CAM & AMT)

May- 2018

ME-757/710.02 Industrial Automation and Robotics

Date: 01.05.2018, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer book /drawing sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q – 1 (a) Draw symbol of 5/3 way control valve and explain working of it. **[03]**

(b) Draw symbol of check valve with spring. **[01]**

(c) Draw symbol of 3/2 directional valve. **[01]**

(d) What is automation? Different between industrial automation and industrial robot. **[02]**

Q – 2 (a) Explain procedure of analog to digital conversion in data acquisition system with suitable sketch also explain components used in analog to digital conversion in details. **[06]**

(b) Draw and explain pneumatic circuit to control cylinder using flip flop. **[06]**

OR

Q – 2 (a) Draw and explain pneumatic circuit for damping device having variable damping force. **[06]**

(b) Draw and explain hydraulic circuit for a robot arm. **[06]**

Q – 3 Attempt any four **[16]**

(a) Explain absolute rotary and linear encoder with suitable figure.

(b) What criteria should be considered while selecting the PLC? Also explain advantages and use of PLCs.

(c) Draw the block diagram of microprocessor and explain the characteristics of microprocessor.

(d) Write short note on capacitive proximity sensor.

(e) List down different steps of vision system also explain object recognition in detail.

SECTION – II

Q-4 Answer the following question.

(a) Write down three laws of robot. **[03]**

(b) What is robot kinematics? **[02]**

(c) Derive the matrix for roto- translation along and about z-axis. **[02]**

Q-5 Do following with respect to a 5-DOFIndustrial Manipulator (**Figure 1**) : **[12]**

1. Label joint co-ordinate system,
2. Prepare joint parameter table,
3. Overall direct kinematics transformation matrix.

OR

Q-5 Write down following for a Stanford manipulator (**Figure 2**) : **[12]**

1. Schematic diagram and joint co-ordinate system,
2. Table of joint parameter,
3. Overall direct kinematics transformation matrix.

Q-6 Attempt any Four **[16]**

(a) Explain relation between the two link axes of robot.

(b) Describe working principle of DC servomotor with layout of drive.

(c) Enlist gripper design considerations.

(d) Explain rectangle coordinate arm geometry of robot including its work envelop, advantages and disadvantages with suitable figure.

(e) Explain synchros and resolver in detail with figure.

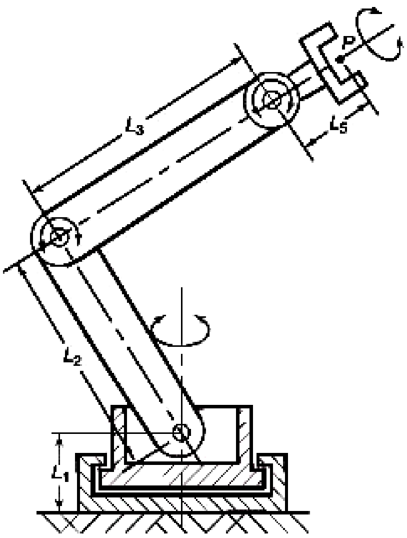

Figure 1

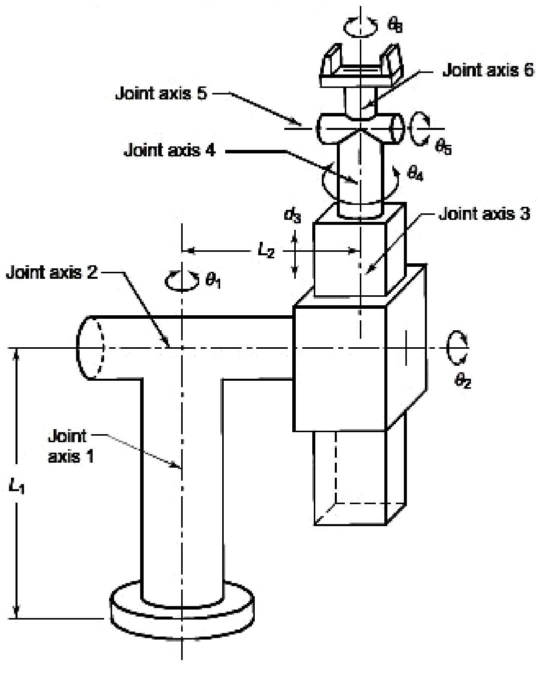

Figure 2

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